



CSE 598 008 EECS 498 003 Lab 2

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Based on slides by Ivan Wei (Fall 2025).

Reminder

- Problem Set 1 due September 25, 11:59pm
 - Starter code in the GitHub repo
 - Includes both chapter 1 and 2!
 - 25 exercises!
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- <https://github.com/GLaDOS-Michigan/eecs498-f26.git>
 - `git clone git@github.com:GLaDOS-Michigan/eecs498-f26.git`

Lab Problems

- Work through exercises in `lab02.dfy` (in the GitHub repo)
- We'll stop and discuss them one by one
- You can find Dafny syntax information on Lecture 3 and 4 slides

Extra - Induction Proof

- To prove a statement involving a natural number n (that is, an integer $n \geq 0$ or 1) holds for all values of n :
 - The **base case** (or **initial case**): prove that the statement holds for 0 , or 1 . (**$P(0)$**)
 - The **induction step** (or **inductive step**): prove that for every n , if the statement holds for n , then it holds for $n + 1$. In other words, assume that the statement holds for some arbitrary natural number n , and prove that the statement holds for $n + 1$. (**$P(n) \Rightarrow P(n + 1)$**)
- The hypothesis in the induction step, that the statement holds for a particular n , is called the **inductive hypothesis**. To prove the induction step, one assumes the induction hypothesis for n and then uses this assumption to prove that the statement holds for $n + 1$.

Extra - Truth Table

Truth tables for basic connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$Q \rightarrow P$	$P \leftrightarrow Q$
f	f	t	f	f	t	t	t
f	t	t	f	t	t	f	f
t	f	f	f	t	f	t	f
t	t	f	t	t	t	t	t